Collaborative Multimodal Learning for Human-Human Interaction Recognition in Videos

Ngoc Thanh Huynh Vo[1,2*], Ngoc Ha Nguyen Thi[1,2*], Tien-Dung Mai[1,2**]

[1] University of Information Technology, Ho Chi Minh City, Vietnam
[2] Vietnam National University, Ho Chi Minh City, Vietnam
21520449@gm.uit.edu.vn, 21520217@gm.uit.edu.vn, dungmt@uit.edu.vn

Abstract. Recognizing the interaction of people is a basic but challenging task in video understanding. In this paper, we propose a new task in this field, named Human-Human Interaction Recognition (HHIR). Our research focuses on recognizing interactions between two individuals in a scene, particularly those involving communication. It is different from the traditional action classification task. HHIR requires modeling complex and multimodal contextual cues across extended video scenes. To address this challenge, we propose a collaborative multimodal learning framework that integrates visual, textual, and audio modalities through an enhanced cross-attention mechanism. Our method leverages ViViT model for spatiotemporal visual feature extraction instead of traditional CNN models, BERT for dialogue understanding, and AST for audio feature extraction. Furthermore, we incorporate a collaborative knowledge integration strategy to enhance cross-modal interactions and apply class weighting to mitigate data imbalance issues. We build a new dataset, constructed from 12 diverse movies, capturing interactions between two individuals instead of isolated actions. Experimental results on this dataset demonstrate that our model achieves 78.54% Micro-F1 and 62.47% Macro-F1, outperforming state-of-the-art methods in multimodal human interaction recognition. This demonstrates the effectiveness of collaborative multimodal learning in capturing complex human interactions. The source code is available at github/hvngocthanh-CS/HHIR.

Keywords: Collaborative multimodal learning · Multitask learning · Multi-label classification · Vision transformers · Interaction recognition · LSTM

1 Introduction

In the digital era, images and videos are essential for information transmission. Recognizing human interactions in videos helps understand context, predict behaviors, and supports applications like security surveillance, violence detection, and action-aware character retrieval. Action Classification and Human-Object

* These authors contributed equally to this work.
** Corresponding author.

Interaction Detection perform well on clear actions (e.g., jogging, shaking hands) but struggle with complex communication-based interactions (e.g., yelling at, asking, giving to, watching) due to unclear action boundaries.

Recently, the Deep Video Understanding (DVU) task has emerged, focusing on interaction prediction in scene-level clips [14, 15]. Existing studies mainly detect a single interaction in short clips (e.g., Kinetics, ActivityNet) [16, 17], while DVU requires predicting multiple interactions in longer scenes.

The DVU challenge in TRECVID 2023 tackles both movie-level and scene-level tasks, with our work being a subproblem of **Finding next or previous interaction**. We aim to develop a model for detecting and recognizing common communication-based human interactions (e.g., talking, asking, yelling) between two individuals. Instead of recognizing isolated actions, our approach captures social and contextual relationships through *multimodal learning*, integrating video, transcribed speech, and audio. Dosovitskiy et al. [8] showed that *Transformers can be effectively applied to vision tasks*. Therefore, we employ ViViT for video feature extraction and leverage powerful pre-trained models like BERT and AST for text and audio.

The remainder of this paper is organized as follows: Section 2 reviews related work, including video understanding, human action recognition, and previous approaches to human interaction recognition. Section 3 clearly defines the HHIR problem and presents our proposed method and framework in detail. Section 4 describes the dataset construction, details the preprocessing and feature extraction processes, and analyzes the experimental results. Finally, Section 5 concludes the study by summarizing the key findings and proposing future directions to advance HHIR and multimodal learning.

2 Related Work

Video understanding has evolved from person clustering and identification [9–11] to story analysis. We review multimodal approaches categorized into early, late, and hybrid fusion [18, 19]. Early fusion combines features before prediction [20], while late fusion aggregates modalities after individual predictions [13]. Although model-agnostic methods are easy to implement, they are generally less effective in extracting multimodal insights. We employ hybrid fusion with cross-attention to better capture relationships between modalities, enhancing interaction context understanding.

2.1 Video Understanding

Deep models facilitate video classification by assigning categories to videos. Benchmarks for activity classification include Kinetics-700, ActivityNet, HACS and UCF-101 [21–24]. The task is evaluated using Top-K accuracy. Recognizing emotions is crucial for understanding human interactions, with Pairwise Emotional Relationship Recognition [7] being a related task.

2.2 Human Action Recognition (HAR)

Turaga et al. [28] distinguished "actions" (individual motions) from "activities" (coordinated interactions) and reviewed recognition methods. Poppe [29], and Presti et al. [32] surveyed action representation and classification. Vrigkas et al. [31] categorized methods into "unimodal" and "multimodal", while Popoola and Wang [30] focused on abnormal behavior detection. Recently, deep learning models such as CNNs, RNN-LSTMs have advanced HAR.

2.3 Human Interaction Recognition (HIR)

Earlier works treat human-human interaction (HHI) as image-level categorical classification [25, 26], while Alper et al. [4] frame it as an image captioning task with free-text targets to capture its contextual nature. Though these tasks involve dynamic images, their methodologies vary, whereas the approach in [1] provides a more generalizable framework.
Unlike Human Action Recognition (HAR), Human Interaction Recognition analyzes interactions between two people. The study in [3] used multi-camera recordings and a fine-tuned ResNet to classify five explicit interactions, achieving 95.45% accuracy. In contrast, our research focuses on communicative interactions, requiring a model integrating visual, audio, and contextual features, along with a tailored dataset.

3 Method

3.1 Problem Statement

The problem of Human-Human Interaction Recognition (HHIR) at the scene level in videos is formulated as a multi-label classification task. The input and output are illustrated in Fig. 1 and defined as follows:

Input:

- A dataset $D = \{(S_i, Y_i)\}_{i=1}^{n}$, where n is the total number of scenes, S_i is the i-th scene in the dataset, and $Y_i \subseteq C$ is the associated set of interaction labels for scene S_i. The set of possible interactions is given by: $C = \{$ talks to, asks, gives to, walks with, watches, yells at, no_interaction$\}$.
- An unseen scene S, which is not part of the training dataset, for which the goal is to predict its set of interactions.

Output: The predicted set of interactions $Y \subseteq C$ that are present in the given scene S.

A scene is a video segment with continuous and coherent events in a specific setting or involving certain characters. **A shot** is the smallest unit in filmmaking, captured as a continuous sequence from the moment recording starts until it stops, without any cuts. A scene consists of multiple shots, and each shot may contain one or more interactions. An interaction often spans multiple shots, so merging them into **a clip** provides better context.

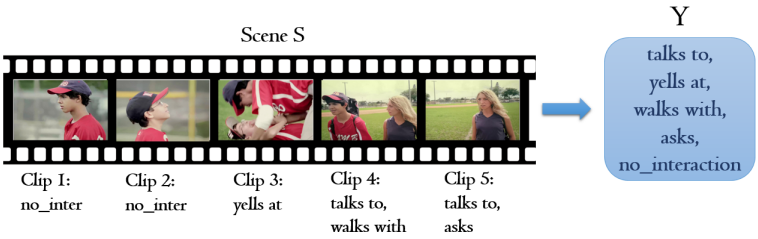


Fig. 1: Illustration of the Human-Human Interaction Recognition task in videos

3.2 Our framework

The HHIR task is a multi-label classification problem at the scene level, where a scene may have multiple interactions. We propose a model that integrates video, text, and audio to classify these interactions. Our framework is illustrated in Figure 2. After extracting feature vectors for the clips, we apply zero-padding to ensure that all scenes have a uniform input size, facilitating stable training and inference while preserving essential information.

Each modality is processed through separate bidirectional LSTMs, each with an output size of hidden_dim×2. This allows the model to capture sequential dependencies and contextual relationships between consecutive clips within a scene, leveraging both past and future information to better comprehend a long-duration event in its entirety.

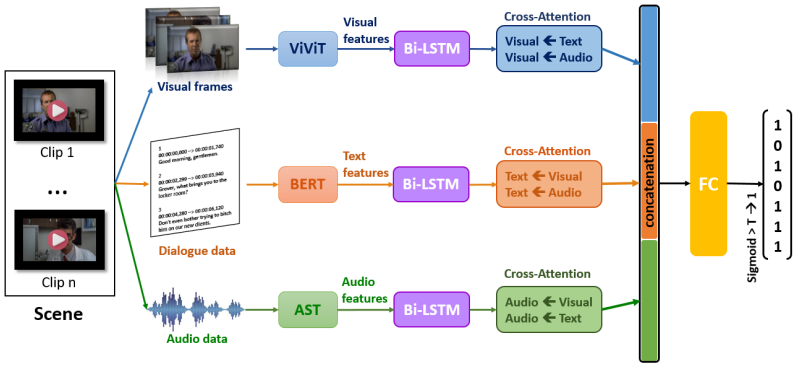


Fig. 2: Pipeline of the proposed method

After processing signals from the three sources (visual, text, audio) through separate LSTMs, the modalities are processed using a parallel cross-attention mechanism with a synergistic effect. Instead of applying cross-attention sequentially to each modality pair (visual-text, visual-audio, text-audio), each modality is updated simultaneously based on interactions with the other two. Specifically, visual features are sequentially refined through cross-attention (Equation 1), first with text features, and then the output is further processed through cross-attention with audio features (collaborative learning). Similarly, text and audio

features are updated based on their interactions with the other two modalities following the same mechanism.

$$\text{Cross-Attention}(Q_1, K_2, V_2) = softmax\left(\frac{Q_1 K_2^T}{\sqrt{d}}\right) V_2 \tag{1}$$

where Q_1 is the query vector from modality 1 (e.g., visual), K_2 and V_2 are the key and value vectors from modality 2 (e.g., text or audio), and d is the dimension of the key vector.

Let v, t, a represent the features of visual, text, and audio, respectively. Then, $v',\ v'',\ t',\ t'',\ a',\ a''$ are the results after each Cross-Attention step. Applying Cross-Attention for each pair of modalities:

$$\begin{aligned} v' &= \text{Cross-Attention}(Q_v, K_t, V_t), \quad & v'' &= \text{Cross-Attention}(Q_{v'}, K_a, V_a) \\ t' &= \text{Cross-Attention}(Q_t, K_a, V_a), \quad & t'' &= \text{Cross-Attention}(Q_{t'}, K_v, V_v) \\ a' &= \text{Cross-Attention}(Q_a, K_v, V_v), \quad & a'' &= \text{Cross-Attention}(Q_{a'}, K_t, V_t) \end{aligned}$$

This approach enables the model to effectively integrate information from multiple sources, preventing bias toward a single modality and strengthening cross-modal relationships. This refined Cross-Attention mechanism allows one modality to indirectly influence another (e.g., visual can impact audio through text). This is particularly important for our task. To preserve the original information and prevent data loss during deep processing, the output of each modality after cross-attention is summed with its initial input (residual connection) before undergoing LayerNorm. Additionally, dropout after attention helps prevent overfitting by randomly masking part of the output during training.

After cross-attention, global average pooling is applied to each modality to aggregate temporal information. The pooled features are concatenated along the last dimension, resulting in a single feature vector of size hidden_dim$\times$6 and passed through a three-layer MLP with ReLU activation, LayerNorm, and Dropout. ReLU and LayerNorm help retain meaningful features and capture complex relationships, while Dropout mitigates overfitting during training. Finally, a linear layer maps the fused representation to the label space for classification. With this design, the model not only learns individual modality-specific representations but also effectively captures the complex relationships among them, thereby improving the accuracy of action classification in videos. The results are passed through a *sigmoid* function to obtain probabilities, which are then compared against a fixed threshold T to determine the presence of interactions in the scene.

3.3 Visual feature extraction for each clip in the scene

Inspired by ViT [12], the Video Vision Transformer (ViViT), a fully Transformer-based model for video classification, was introduced in [8]. To extract visual features, we utilized the ViViT architecture. Specifically, we employed its **Factorized Self-Attention** variant, which reduces computational complexity by

separating self-attention into spatial and temporal components, optimizing efficiency while effectively modeling spatiotemporal interactions. The result is a set of feature vectors, each with a length of 1024.

3.4 Text feature extraction for each clip in the scene

Text features provide essential context, complementing visual features in interaction recognition. For instance, phrases like "Here is yours" or "This is yours" are often associated with the act of handing something to another person. We use a pre-trained BERT model for text feature extraction. Unlike previous language models that consider one-sided context, BERT captures semantic representations by leveraging bidirectional context. As a result, we obtain a set of feature vectors, each with a fixed dimension of 768. This aligns with BERT Base's representation space for encoding input text semantics.

3.5 Audio feature extraction for each clip in the scene

Audio features capture intonation and emotion in speech, aiding interaction recognition. For example, "yells at" is linked to high volume and fast speech rate, indicating anger or tension. We extract audio features using the pre-trained Audio Spectrogram Transformer (AST), which leverages attention mechanisms to process spectrograms and capture long-range context. This results in feature vectors of length 768.

4 Experiments

4.1 Dataset Construction

DVU Dataset: The Deep Video Understanding task requires advanced computer vision to analyze relationships between entities in a video. To facilitate research, the DVU dataset was introduced in the TRECVID-DVU Challenge 2023.

The DVU dataset supports multiple tasks in the DVU Challenge, but this study focuses on three main data types:

1. Scene data from movies.
2. Knowledge Graph data (Fig 5): a collection of JSON files representing relationships between entities in each scene. In this study, we focus on two key entity types: Person (characters in the scene), Interaction (interactions between characters).
3. Face identification data (Fig 6): used to recognize and assign names to characters based on their faces in the movie.

We experimented on 12 movies, selected from a total of 19 movies in the DVU training set, splitting scenes into train, validation, and test sets at a 6:2:2

ratio. These scenes were divided into 11,386 shots, after labeling and merging, resulted in 9,113 clips. Details are shown in Table 1.

Table 1: Details of the 12 movies in the experimental dataset

Movie	**Genre**	**Length (min)**
Honey	Romance	86
The Big Something	Comedy	101
Time Expired	Comedy / Drama	92
Valkaama	Adventure	93
Bagman	Drama / Thriller	107
Road to Bali	Comedy / Musical	90
The Illusionist	Adventure / Drama	109
Calloused Hands	Drama	92
Chained For Life	Comedy / Drama	88
Liberty Kid	Drama	88
Huckleberry Finn	Adventure	106
Losing Ground	Comedy / Drama	81

Video Dataset Construction: A shot labeled as "no_interaction" if no interaction occurs. Without shot segmentation, the model may struggle to distinguish interaction from non-interaction segments. Dividing a scene into shots ensures more intuitive annotation and helps the model focus on relevant time intervals, enhancing interaction recognition performance, Fig (4).

Shot	File	Label
Shot 1	shot_a.mp4	talks to
Shot 2	shot_a.mp4	walks with

Fig. 3: Illustration of labeling when a shot contains multiple simultaneous interactions

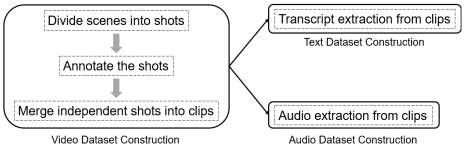


Fig. 4: Dataset Construction Process

1. Divide scenes into shots We used the PySceneDetect library [3] to segment scenes into shots. This library supports multiple detection algorithms, and we utilize ContentDetector to identify significant frame changes. The threshold value, ranging from 0 to 100, is adjusted based on scene characteristics and represents grayscale pixel differences between consecutive frames.

- Low threshold: more sensitive, detecting a greater number of shots but potentially causing false splits due to noise or minor movements.
- High threshold: less sensitive, detecting only significant scene changes, which helps reduce false splits but may miss shorter shots.

2. Shot Annotation Each JSON file in the Knowledge Graphs dataset provides information about interactions occurring within a scene but does not specify which interactions take place in individual shots. Therefore, after segmenting the scene into shots, we perform manual annotation by analyzing each shot within the scene. This process integrates information from Knowledge Graphs

[3] www.scenedetect.com

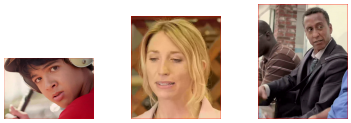
and face identification data to accurately assign labels to each shot. Refer to the interaction label definitions this link.

In Figure 5, each node contains three attributes: `text` (entity name), `key` (entity key, a unique integer used to identify the node in the graph), `type` (entity type). Note that this study focuses only on entities with type is Interaction and Person. Next, each link represents a connection between two nodes and has two attributes: `from` (the key of the source entity where the link starts), `to` (the key of the target entity where the link points).

Nodes		
Text	Key	Type
Josh	-2	Person
Rabbi Brookstein	-3	Person
Talks to	-4	Interaction

Links	
From	To
-3	-4
-4	-2

Fig. 5: Illustration of Knowledge Graphs data of a scene



(a) Josh1 (b) Debbie1 (c) Byrd1

Fig. 6: Face identification images of characters in a movie.

To determine interaction labels, we first identify the key of the *Interaction* node and use it to find linked entities. If both linked entities are of type *Person*, we extract their names from the text; otherwise, the interaction is ignored. For example, "talks to" has key -4, linking entities -3 and -2, corresponding to *Rabbi Brookstein talks to Josh*. We then match character names with face identification data, analyze shots to identify interactions, and store the results in a CSV file. For communication interactions, analyzing character dialogues is essential. We extract text data for the entire scene, as in 4.1, to accurately assign labels.

3. Merge independent shots into clip

An interaction often spans multiple consecutive shots. Therefore, merging these shots into a clip ensures a more comprehensive representation of the interaction's context. After completing the annotation process, we used the MoviePy library [4], the `concatenate_videoclips` function used to merge consecutive shots with the same interaction label into complete clips. We apply a data overlap technique to ensure continuity between clips, where each subsequent clip repeats a portion of the previous one. This approach preserves interaction context and minimizes information loss during segmentation. The complete procedure for shot-to-clip segmentation is detailed in Algorithm 1. We set $L = 3$, so each clip contains up to three consecutive shots with the same interaction label. This value offers a good balance between context coverage and computational efficiency. L can be further adjusted to meet different application needs.

Text Dataset Construction: To transcribe video content with multilingual support, we use the SubRip Subtitle (SRT) format, which provides speech seg-

[4] https://zulko.github.io/moviepy/

Algorithm 1: Shot-to-Clip Segmentation

Input: List of shots with interaction labels;
L: maximum number of shots per clip;
k: step size (distance between the starting indices of consecutive clips)
Output: List of clips (each clip is a sequence of consecutive shots, with its label)
Group shots by their interaction label. For each label c, denote the sequence of shots as $S_c = (s_1, s_2, ..., s_{n_c})$, where n_c is the number of shots with label c;
foreach *label c, sequence* S_c **do**
  for $i \leftarrow 1$ **to** n_c k **do**
    $start \leftarrow i$, $end \leftarrow \min(i + L - 1, n_c)$
    $clip \leftarrow (s_{start}, s_{start+1}, ..., s_{end})$
    Save information of *clip* together with label c to the result list;
Concatenate shots into clips, and write the clip information to a CSV file.

ments with precise timestamps for accurate synchronization. We deploy Whisper Large-v3 via Groq API [5] to extract dialogues in SRT format from collected clips. Groq API offers an optimized platform for high-performance AI models. Whisper, developed by OpenAI, is an advanced Automatic Speech Recognition (ASR) model trained on large-scale data, enabling accurate multilingual speech recognition. Compared to previous versions, Whisper Large-v3 delivers higher accuracy and improved performance, especially in multilingual environments.

Audio Dataset Construction: Extracting audio from video is a crucial step in building audio datasets for processing and analysis tasks. FFmpeg, a powerful open-source framework, is widely used due to its extensive format support and cross-platform compatibility. With its flexibility and extensive format support, FFmpeg is a powerful tool for extracting audio data from videos, effectively meeting the requirements of research and real-world applications.

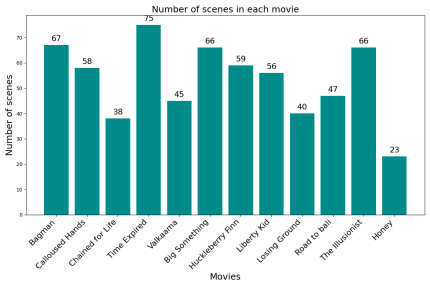


Fig. 7: The number of scenes in the 12 movies

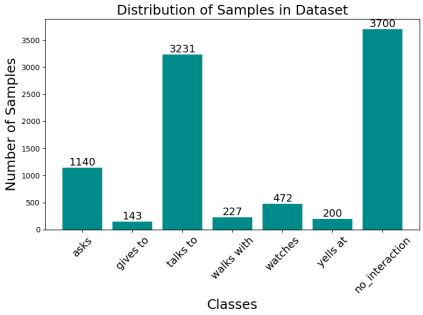


Fig. 8: Distribution of the number of clips across labels

4.2 Preprocessing

- **Video preprocessing:** use OpenCV to read each frame from the video, resize it to (128,128) using `cv2.resize`, normalize to the range [0,1], and convert it into a tensor for model input.
- **Dialogue transcript preprocessing:** use `pysrt` to read the SRT file, extract and clean the text by removing timestamps, bracketed content, and extra spaces. Finally, normalize sentence length by adding `<PAD>` to shorter sentences.
- **Label set preprocessing:** use one-hot encoding to transform the labels into a binary vector indicating the presence of each label in a clip. The set of labels for a scene is determined by the union of labels appearing in its clips.

4.3 Feature Extraction Process

- ViViT model initialization: `image_size=(128x128), num_frames=512, ImgPatchSize=16, FramePatchSize=2, spatial_depth=6, temporal_depth=6, heads=8, variant="factorized_self_attention"`.
- Initialize `AutoTokenizer` and `AutoModel` from transformers to tokenize text and load the pre-trained BERT-base model. The pre-trained BERT-base model then extracts features from the [CLS] token.
- Load the pre-trained AST model and `AutoFeatureExtractor` to process raw audio into features like mel-spectrograms. Convert audio to 16kHz, apply padding, and extract features from `last_hidden_state` without classification.

4.4 Training Details

We initialize the parameters for the proposed model: `hidden_dim = 512`: the hidden state dimension, representing the number of neurons in each LSTM layer, `num_labels=7, dropout=0.3, num_heads_attn=8, dropout_attn=0.4`.

We propose using the BCE loss, implemented in PyTorch as `BCEWithLogitsLoss()`, treating each label as an independent binary classification task and averaging the loss across all labels. Due to the imbalance in class distribution, we propose applying class weights (`pos_weight`) to the loss function to encourage the model to focus more on rare labels:

$$\text{class_weights}_i = \frac{\text{total_samples} - \text{class_count}_i}{\text{class_count}_i} \tag{2}$$

- class_count_i: the number of samples in class i.
- total_samples: the total number of samples in the dataset.

We utilize the AdamW [27] optimizer with a learning rate of 3e-5 and a weight decay of 1e-5 and the batch size is 16. Training is conducted for 50 epochs at three threshold levels $T = \{0.5, 0.6, 0.7\}$ on a P100 GPU.

4.5 Results

Experimental Results To evaluate the performance of HHIR, we use Macro-F1 and Micro-F1, considering the imbalanced categories. Macro-F1 is the arithmetic mean of the per-class F1-scores, which gives equal weights to each class. Micro-F1 is the harmonic mean of precision and recall calculated over all the samples without considering the class weights. Additionally, we conduct experiments on all three probability thresholds T, where an interaction is considered present in the scene if the model's predicted probability exceeds T (Modality: V-Visual, T-Text, A-Audio).

Table 2: Evaluation of different modality combinations on the test set using *the typical cross-attention architecture*

Modality	Metric	Without class weights			With class weights		
		T=0.5	T=0.6	T=0.7	T=0.5	T=0.6	T=0.7
V-A	Macro-F1	0.5069	0.4904	0.4750	0.5525	0.5586	0.5523
	Micro-F1	0.7490	0.7456	0.7418	0.7288	0.7337	0.7339
V-T	Macro-F1	0.5254	0.5237	0.4810	0.5855	0.5878	0.5741
	Micro-F1	0.7823	0.7867	0.7797	0.7651	0.7718	**0.7742**
V-T-A	Macro-F1	0.5649	**0.5711**	0.5643	**0.5988**	0.5831	0.5790
	Micro-F1	0.7837	0.7870	**0.7918**	0.7668	0.7662	0.7720

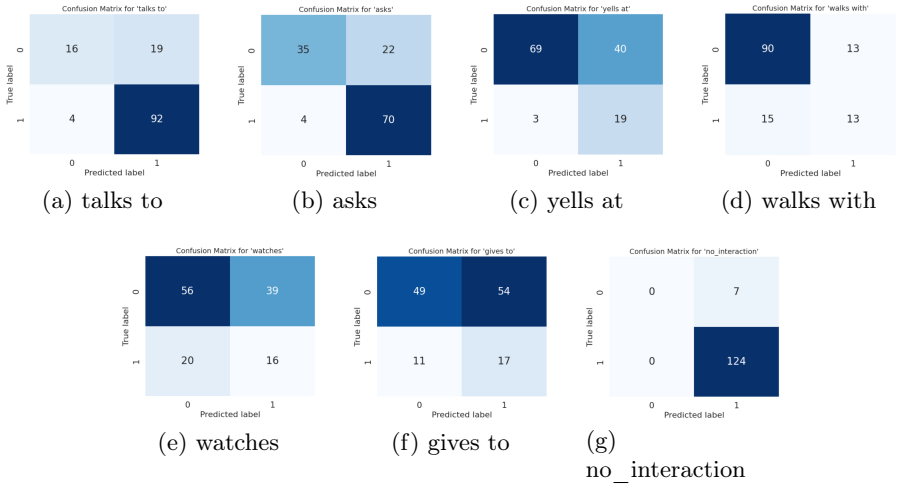
Table 3: Evaluation of different modality combinations on the test set using *the refined cross-attention architecture*

Modality	Metric	Without class weights			With class weights		
		T=0.5	T=0.6	T=0.7	T=0.5	T=0.6	T=0.7
V-A	Macro-F1	0.4957	0.4752	0.4429	0.5026	0.4856	0.4724
	Micro-F1	0.7500	0.7451	0.7373	0.7223	0.7224	0.7240
V-T	Macro-F1	0.5380	0.5259	0.5313	0.6046	0.6042	0.5835
	Micro-F1	0.7724	0.7715	0.7817	0.7556	0.7611	0.7568
V-T-A	Macro-F1	**0.5566**	0.5469	0.5247	0.6195	**0.6247**	0.6077
	Micro-F1	0.7970	**0.7995**	0.7974	0.7757	0.7854	**0.7904**

Most studies on human interaction recognition are formulated as single-label classification tasks and evaluated using *Accuracy*. However, our problem is multi-label, which is a relatively novel, requiring a more suitable evaluation approach. To ensure a fair comparison, we benchmark our results against the Pairwise Emotional Relationship Recognition (PERR) task, which also adopts a multi-modal approach with three modalities: visual, text, and audio, and is evaluated using the same metrics.

Analysis

- Observing the results in Table 3, the model that integrates all three modalities outperforms the one that combines only two modalities when evaluated at the same probability threshold.



(a) talks to (b) asks (c) yells at (d) walks with

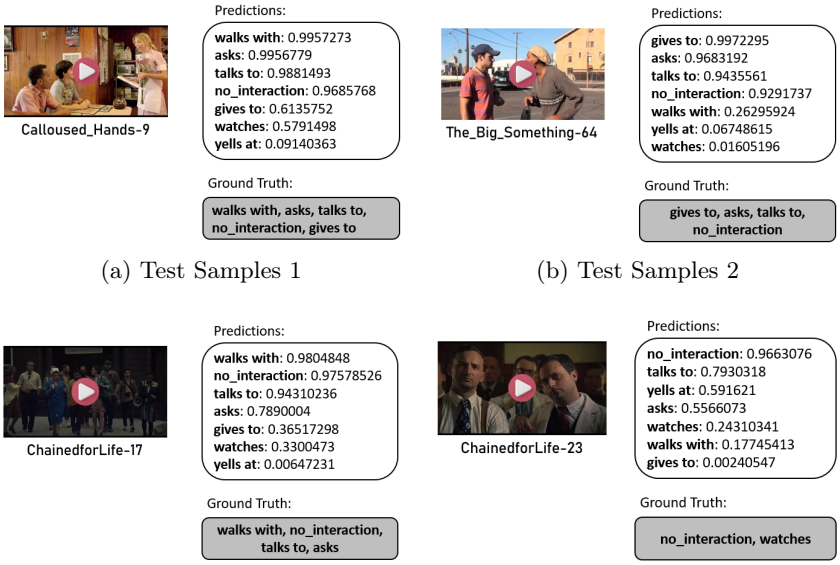
(e) watches (f) gives to (g) no_interaction

Fig. 9: Confusion matrices computed at the scene level using class_weights from V-T-A with a threshold 0.6

Table 4: Comparison of two different tasks using multimodal features in terms of Micro-F1 and Macro-F1

Task	Method	Visual-Text-Audio	
		Micro-F1	Macro-F1
PERR [7]	$\text{SMTA}_{\text{Transformer}}$	0.7005	0.6173
$\text{HHIR}_{\text{T}=0.5}$ (ours)	LSTM+Cross-attention	0.7757	0.6195
$\text{HHIR}_{\text{T}=0.6}$ (ours)	LSTM+Cross-attention	0.7854	**0.6247**
$\text{HHIR}_{\text{T}=0.7}$ (ours)	LSTM+Cross-attention	**0.7904**	0.6077

- Audio plays a crucial role in enhancing the model's performance, especially for the *asks* and *yells at* interactions, which heavily rely on acoustic information.
- The model without pos_weight has a higher F1-score micro as it focuses on common labels. In contrast, the model with pos_weight has a lower F1-score micro but a higher F1-score macro. Using pos_weight helps the model learn rare labels better but may trade off overall performance.
- When applying class weighting, most values in Table 3 exceed those in Table 2, especially for V-T-A, indicating that the refined cross-attention model integrates information more effectively. The higher F1-Macro score demonstrates improved recognition of rare interactions, mitigating class imbalance.
- In our study, we experiment with thresholds starting from 0.5, as F1-macro tends to reach its peak from this value onward. The choice of threshold can be adjusted based on computational resources and specific task requirements to achieve optimal performance.



(a) Test Samples 1 (b) Test Samples 2

(c) Test Samples 3 (d) Test Samples 4

Fig. 10: The image shows prediction probabilities for some scenes in the test set

5 Conclusion

In this paper, we propose a new task named Human-Human Interaction Recognition as well as a new dataset. Refining cross-attention with a collaborative knowledge integration approach enhances interaction recognition ability, especially for rare labels, compared to the traditional design. Our method outperforms the SMTA [7] method in both F1-micro and F1-macro scores, Table 4. However, the noticeable gap between them suggests that while effective overall, it still needs improvement in handling rare labels. The primary reason is the severe class imbalance in the dataset, Fig 8. In multi-label classification, reducing samples containing common labels can affect the distribution of other labels, posing a significant challenge in our research. Merging shots into overlapping clips enhance data augmentation naturally and reduce information loss at clip boundaries but increases computational cost, redundancy, and less relevant information. Despite differing datasets and tasks, both PERR and HHIR integrate multi-modal information (video, text, audio) for video analysis, making their comparison valid. Our findings provide a comprehensive perspective on this emerging field.

Although multimodal large language models (multimodal LLMs) show strong potential in unified processing and integration of multimodal data, they often require substantial computational resources, making them unsuitable for our research context. Instead, we adopt a strategy that processes each modality separately and integrates them via cross-attention, allowing for flexible optimization

on a single P100 GPU. This approach effectively leverages state-of-the-art pre-trained models for each modality, which is particularly advantageous given the limited data and the specific requirements of the HHIR task. In future work, we plan to explore the integration of multimodal LLMs to compare performance and assess their potential for improving model generalization. We expect the HHIR task can inspire more insights in behavior analysis, video understanding, and multimodal learning.

Acknowledgments. This research is funded by Vietnam National University Ho Chi Minh City (VNU-HCM) under grant number C2024-26-06.

References

1. Pruss, S., Alper, M., Averbuch-Elor, H.: Dynamic Scene Understanding from Vision-Language Representations. arXiv:2501.11653 (2025).
2. Shaikh, M.B., et al.: From CNNs to Transformers in Multimodal Human Action Recognition: A Survey. arXiv:2405.15813 (2024).
3. Khean, V., et al.: Human Interaction Recognition in Surveillance Videos Using Hybrid Deep Learning and Machine Learning Models. Comput. Mater. Contin. 81(1), 773–787 (2024).
4. Alper, M., Averbuch-Elor, H.: Learning Human-Human Interactions in Images from Weak Textual Supervision. In: ICCV, pp. 12345–12354. IEEE (2023).
5. Radford, A., et al.: Robust Speech Recognition via Large-Scale Weak Supervision. arXiv:2212.04356 (2022).
6. Gabeur, V.: Multi-Modal Learning for Video Understanding. Ph.D. thesis, Université Grenoble Alpes (2022).
7. Gao, X., et al.: Pairwise Emotional Relationship Recognition in Drama Videos: Dataset and Benchmark. arXiv:2109.11243 (2021).
8. Arnab, A., et al.: ViViT: A Video Vision Transformer. arXiv:2103.15691 (2021).
9. Brown, A., et al.: Automated Video Labelling: Identifying Faces by Corroborative Evidence. In: MIPR, pp. 175–184. IEEE (2021).
10. Brown, A., et al.: Face, Body, Voice: Video Person-Clustering with Multiple Modalities. In: ICCVW, pp. 2698–2707. IEEE (2021).
11. Kalogeiton, V., Zisserman, A.: Constrained Video Face Clustering Using 1NN Relations. In: BMVC. pp. 1–12. BMVA Press (2020).
12. Dosovitskiy, A., et al.: An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale. arXiv:2010.11929 (2020).
13. Kukleva, A., et al.: Learning Interactions and Relationships between Movie Characters. In: CVPR, pp. 10189–10198. IEEE (2020).
14. Carreira, J., Zisserman, A.: Quo Vadis, Action Recognition? In: CVPR, pp. 6299–6308. IEEE (2017).
15. Yan, S., Xiong, Y., Lin, D.: Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In: AAAI Conference on Artificial Intelligence (AAAI), pp. 7444–7452. AAAI Press (2018).
16. Kay, W., et al.: The Kinetics Human Action Video Dataset. arXiv preprint 1705.06950 (2017).
17. Heilbron, F.C., et al.: ActivityNet: A Large-Scale Video Benchmark for Human Activity Understanding. In: CVPR, pp. 961–970. IEEE (2015).

18. D'Mello, S.K., Kory, J.: A Review and Meta-Analysis of Multimodal Affect Detection Systems. ACM Comput. Surv. 47(3), 1–36 (2015).
19. Atrey, P.K., et al.: Multimodal Fusion for Multimedia Analysis: A Survey. Multimed. Syst. 16(6), 345–379 (2010).
20. Castellano, G., et al.: Emotion Recognition through Multiple Modalities: Face, Body Gesture, Speech. In: Affect and Emotion in Human-Computer Interaction, pp. 92–103. Springer (2008).
21. Carreira, J., et al.: A Short Note on the Kinetics-700 Human Action Dataset. arXiv:1907.06987 (2019).
22. Heilbron, F.C., et al.: ActivityNet: A Large-Scale Video Benchmark for Human Activity Understanding. In: CVPR, pp. 961–970. IEEE (2015).
23. Zhao, H., et Íal.: HACS: Human Action Clips and Segments Dataset for Recognition and Temporal Localization. In: ICCV, pp. 8668–8678. IEEE (2019).
24. Soomro, K., et al.: UCF101: A Dataset of 101 Human Action Classes from Videos in the Wild. arXiv:1212.0402 (2012).
25. Xiong, Y., et al.: Recognizing Complex Events from Static Images by Fusing Deep Channels. In: CVPR, pp. 1600–1609. IEEE (2015).
26. Antol, S., et al.: Zero-Shot Learning via Visual Abstraction. In: ECCV, pp. 401–416. Springer (2014).
27. Loshchilov, I., Hutter, F.: Decoupled Weight Decay Regularization. arXiv:1711.05101 (2017).
28. Turaga, P., et al.: Machine Recognition of Human Activities: A Survey. IEEE Trans. Circuits Syst. Video Technol. 18(11), 1473–1488 (2008).
29. Poppe, R.: A Survey on Vision-Based Human Action Recognition. Image Vis. Comput. 28(6), 976–990 (2010).
30. Popoola, O.P., Wang, K.: Video-Based Abnormal Human Behavior Recognition: A Review. IEEE Trans. Syst. Man Cybern. C 42(6), 865–878 (2012).
31. Vrigkas, M., Nikou, C., Kakadiaris, I.A.: A Review of Human Activity Recognition Methods. Front. Robot. AI 2, 28 (2015).
32. Lo Presti, L., La Cascia, M.: 3D Skeleton-Based Human Action Classification. Pattern Recognit. 53, 130–147 (2016).
33. Vaswani, A., et al.: Attention is All You Need. arXiv:1706.03762 (2017).
34. Shu, X., Tang, J., Qi, G.J., Liu, W., Yang, J.: Hierarchical Long Short-Term Concurrent Memory for Human Interaction Recognition. arXiv:1811.00270 (2018).